SEGi UNIVERSITY

BCC1313 – ALGORITHMS & COMPLEXITY

Tutorial 4

1. Define a stack. If you push the letters A, B, C and D in order onto a stack of characters and then pop them, in what order will they be deleted from the stack?

2. What do initially empty stack *Stack1* and *Stack2* "look like" after the following sequence of operations:

   Stack1.push(1)
   Stack1.push(2)
   Stack2.push(3)
   Stack2.push(4)
   Stack1.pop()
   Stack2.getTop(stackTop)
   Stack1.push(stackTop)
   Stack1.push(5)
   Stack2.pop(stackTop)
   Stack2.push(6)

3. For the following input line, trace the execution of *readAnd Correct* and show the contents of the stack at each step:

   abc←de←←fg←h

4. For each of the following strings, trace the execution of the balanced-braces algorithm and show the contents of the stack at each step.
   a) X{{YZ}}}
   b) {X{Y{{Z}}}
   c) {{{X}}}

5. For each of the following strings, trace the execution of the language-recognition algorithm and show the contents of the stack at each step.
   a) XY$ XY
   b) XY$ X
   c) Y$ YX
   d) XX$ XX
   e) XY$ Y

6. Use the stack algorithms to evaluate the postfix expression ab-c+. Assume the following values for the identifiers: a=7, b=3, c=-2. Show the status of the stack after each step.

7. Use the stack algorithms to convert the infix expression a/b*c to postfix form. Be sure to account for left-to-right association. Show the status of the stack after each step.

8. Evaluate the following postfix expressions by using the stack algorithms. Show the status of the stack after each step of the algorithm. Assume the following values for the identifiers: a=7, b=3, c=12, d=-5, e=1.
   a) abc+-
   b) abc-d*+
   c) ab+c-de*+

9. Convert the following infix expressions to postfix form by using stack algorithms. Show the status of the stack after each step of the algorithm.
   a) a-b+c
   b) a/(b*c)
   c) (a+b)*c
   d) a-(b+c)
   e) a-(b+c*d)/e